

TrES-5b

R-0792

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_221024 · created 2026-07-28T07:38:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459876.6496 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.006 +/- 0.053
Transit depth (Rp/Rs)^2	0.0034 +/- 0.061 [%]
Orbital Inclination (inc)	87.69 +/- 2.95
Airmass coefficient 1 (a1)	293.02 +/- 223.02
Airmass coefficient 2 (a2)	-3.0 +/- 0.26
Scatter in the residuals of the lightcurve fit is	80.83 %
Best Comparison Star	None
Optimal Aperture	6.08
Optimal Annulus	8.4
Transit Duration (day)	0.0622 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.006 ± 0.053 vs 0.142 (deviation 1.29σ)
Duration fit vs published	0.0622 d vs 0.0758 d (deviation 0.88σ)
Mid-time O–C	-4.6 min
Depth SNR	0.1
Residual scatter	80.83 %

Light curve

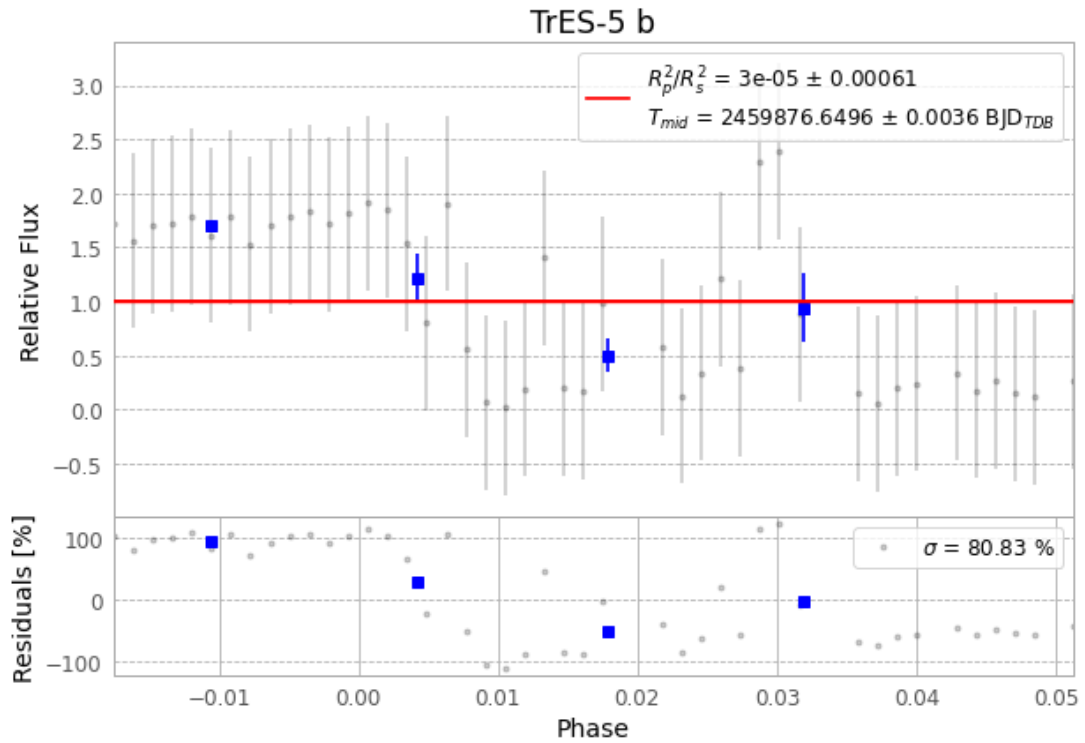


Figure 1 — FinalLightCurve_TrES-5 b_2022-10-23.png (SHA-256 1fa6c5cef693fd99...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a911a091101b98a7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

52 FITS frames (34 MB), TRES-5221024025418.FITS → TRES-5221024052715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-221024.log (11851a7b7b35...), FinalLightCurve_TrES-5 b_2022-10-23.png (1fa6c5cef693...), FinalLightCurve_TrES-5 b_2022-10-23.csv (df1ef5357e00...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 07:38Z.